

HAMMAD AHMED

Full-Stack Developer · AI Integration · Cloud-Deployed Systems+92 304-5291947 · hammadahmed9110@gmail.com · linkedin.com/in/hammadahmed911

PROFESSIONAL SUMMARY

CS student at NUST building production-grade full-stack systems — React/Next.js frontends, FastAPI/Node.js backends, AWS deployments with Docker. Shipped multiple projects spanning AI integration (Gemini, RAG, Pinecone), real-time infrastructure (WebSockets, Redis), distributed systems, and ML. Found and patched 10+ critical vulnerabilities during a cybersecurity internship. Led teams of up to 99 people.

WORK EXPERIENCE

Full-Stack Developer Intern — Ulearn, Lahore

Jun – Aug 2025

- Built React + Node.js dashboards adopted team-wide, cutting ~20% of daily manual work within the first month.
- Replaced a fragile data layer with RESTful APIs, reducing backend error rates and frontend integration time.
- Ran web penetration testing with Nmap and Metasploit — found and patched 10+ critical vulnerabilities before production.

PROJECTS

BookMind AI Next.js · FastAPI · Pinecone · Redis · AWS

- Built semantic PDF search with vector embeddings, Gemini-powered chat, and auto-graded quizzes — shipped in 6 weeks.
- Cut query latency under load with Redis caching + connection pooling; zero-downtime AWS deploy with auto-scaling.

HiveMind Next.js · TypeScript · WebSockets

- Implemented real-time chat, SSR dynamic feeds, JWT auth, and protected routing from scratch.
- Modular architecture with optimistic UI updates — new features plug in without structural changes.

SwiftByte Turborepo · Express · Prisma · PostgreSQL · Socket.io

- Built four role-scoped portals (customer, restaurant, rider, admin) on a shared typed API with full ordering flow.
- Live order status via Socket.io; JWT auth, role-scoped routing, and rate-limited middleware across all apps.

Edge-Cloud Replication gRPC · Raft · HLC · Distributed Systems

- Four-layer replication stack: Raft consensus at edges, causal async updates to cloud via HLC.
- Out-of-order buffer + deduplication; verified RYW, monotonic reads, and post-heal convergence via fault injection.

Android Malware Detection LightGBM · SHAP · Streamlit

- Trained multi-class APK classifier (81 features) across trojan, adware, and ransomware families using LightGBM and Random Forest.
- SHAP attribution per prediction; Streamlit app for APK upload and risk-scored output.

Musify FastAPI · Web Audio API · React · Vite

- Classified instruments from uploaded photos; synthesized timbre in the browser using additive synthesis and PeriodicWave.
- Harmonic extraction from samples, keyboard-triggered playback, and live spectrum visualizer.

EDUCATION

B.S. Computer Science — National University of Science and Technology (NUST)

Sep 2023 – Present

- Coursework: Artificial Intelligence, Information Security, Computer Networks, Data Structures & Algorithms.
- Active in hackathons and coding competitions; leadership roles in student organizations.

LEADERSHIP & VOLUNTEER EXPERIENCE

Deputy Director, Orientation Guides — Orientation of NUST

- Led 99 guides to onboard 4,500 incoming freshmen — coordinated logistics, training, and on-ground execution.

Deputy Director, Internal Affairs — NUST Olympiad

- Managed 45-person ops team delivering a smooth experience for 3,000 participants across concurrent sporting events.

Deputy Director, Marketing — NUST Community Service Club

- Led 30-member team; raised PKR 1.7M for Ramadan rations, collected 1,300 pints of blood, and ran women empowerment drives.

House Lead — Science Bee NUST

- Coordinated student participation and competition logistics for an inter-school science olympiad.

SKILLS & CERTIFICATIONS

Stack: JS/TS, Python, Java, C · React, Next.js, Node.js, FastAPI · MySQL, Redis, Pinecone · Docker, AWS, Prometheus

AI & Security: Gemini AI, RAG pipelines, Semantic Search · Nmap, Metasploit, OWASP, Web Penetration Testing

Certifications & Languages: Cybersecurity · Machine Learning · SEO · Freelancing · English (professional) · Urdu (native)